Claim Denial Triage Agent for Outpatient Revenue Cycle

Students: Neha Kashyap (nkashya6@asu.edu), Hriday Pradhan (hpradha3@asu.edu),
Silas Sangmin (ssangmin@asu.edu), Sachin Kothandaraman (skothan4@asu.edu)
Stakeholder: Mokshitha Mohan (Meeting Date: 4th October, 2026)
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Trace: 1, 2

Section 1. Refined Problem Definition and Scope

Problem: Outpatient practices lose revenue due to denied claims, and a large share of those claims are never reworked. The cause is usually late triage, and by the time someone realises a denial was recoverable, the payer's filing window has closed. The project builds an agent that takes a single denied claim and prepares a recommendation on why it was denied, what should be done about it, and when it should be completed.
Intended user and scenario: The user is an accounts receivable specialist at an outpatient practice working on a denial worklist. The specialist opens one denied CMS-1500 claim. The system prepares a review package recommending whether to correct and resubmit, appeal with documentation, route to another payer, bill the patient, post a contractual adjustment, write off, or escalate for manual review. The specialist reviews the evidence, edits any draft, and makes the final decision. The system will not submit claims or appeals.
Input: One claim record in JSON for a professional (CMS-1500) outpatient claim: payer, dates of service and remittance, service lines with CPT codes and amounts, the CARC and RARC denial codes returned, and encounter facts. Evidence-dependent cases also include a synthetic clinical note. The agent may retrieve supporting information from a local collection of payer policy documents and synthetic payer eligibility, or claim history records implemented for the project.
Output: The system produces a structured record and a readable review report. For each denial or affected service line, the output includes the root-cause category or an unknown result, the recommended disposition, the rationale, the required actions, the evidence used, the missing evidence, the applicable deadline, and the review status. It also identifies the primary claim-level recommendation and records the tool calls and sources used. When an appeal is both allowed and supported by the retrieved policy and encounter evidence, the system produces a draft appeal with policy and chart citations. Otherwise, it explains why drafting was withheld and escalates the claim when necessary.
Tasks the system performs:
The system validates the claim; maps known CARC and RARC combinations using the deterministic baseline; determines what additional information is needed; retrieves the relevant payer profile, policy passages, and available synthetic encounter records; compares policy criteria with documented clinical evidence; identifies missing or conflicting evidence; recommends a disposition; calculates the applicable deadline deterministically; verifies that the proposed action is allowed; and generates a grounded appeal draft only when the evidence supports it. Unmapped codes, insufficient evidence, contradictory sources, expired deadlines, and tool failures result in manual review rather than a decision based on guesswork.
Scope:

- In scope: professional outpatient claims, the baseline's ten known root-cause categories plus unknown/manual-review handling, evidence-aware disposition recommendations, action planning, deterministic deadline calculation, payer-policy retrieval, synthetic clinical-note interpretation, grounded appeal drafting, evaluation, audit traces, and human approval.
- Out of scope: institutional (UB-04) claims, live clearinghouse or EHR integration, automated submission to any payer, legal interpretation of actual payer contracts, and real patient data (everything is synthetic).

Success and failure from the stakeholder's perspective: Success is a call a trained biller would make: the root cause and disposition match a trained reviewer's judgment, the actions identify the actual claim or evidence defect rather than restating the denial, the deadline is right for that payer profile, and the retrieved evidence supports the recommendation. The appeal letter should need no or light editing rather than rewriting, with no invented clinical or policy facts and no unsupported statements. Failure is a wrong root cause, an appeal filed on a contractual adjustment, a missed deadline, a letter that asserts something the chart does not support, an unsupported appeal

statement, failure to escalate insufficient evidence, or any submission that skips the approval gate. The system should escalate when the available evidence is insufficient, contradictory, or uncertain.

Section 2. Stakeholder Discovery and Feedback

Our discussion with Mokshitha Mohan confirmed that the Claim Denial Triage Agent should help accounts receivable specialists determine why a professional outpatient claim was denied, what action to take, and when that action is due. The current baseline validates a synthetic claim, interprets its denial codes, applies payer-specific rules, calculates the deadline, and produces a disposition, action plan, and appeal draft when appropriate. The proposed agent will extend this workflow by retrieving payer policies and reviewing supporting encounter evidence for cases that the rule-based baseline cannot confidently resolve.

Mokshitha emphasised that safety and reliability are the main priorities. She agreed that the system should require manual review when evidence is missing or conflicting, the denial code is unknown, the applicable payer policy cannot be found, the deadline has expired, or a required tool repeatedly fails. The project will continue to use synthetic data, exclude live EHR or clearinghouse integration, and require human approval before any external action. She did not request specific payers or procedures for the initial policy collection, allowing the team to select a feasible representative set.

Her feedback refined both the safety requirements and the definition of success. In addition to unsupported appeals, incorrect deadlines, incorrect payer policies, contractual adjustment appeals, and decisions made without sufficient evidence, she identified unjustified patient billing as a critical safety error. She also stated that missing a deadline due to a tool failure or timeout should be treated as seriously as miscalculating the deadline. Success, therefore, requires more than resolving additional cases: those resolutions must match the labelled answers and contain no critical safety errors.

Mokshitha confirmed the refined project scope and agreed to review approximately 10 to 15 representative synthetic cases or system reports, rating the recommendation, action plan, supporting evidence, and appeal draft. She also recommended testing on more than 40 held-out cases when feasible and reporting errors by decision type, such as confusing resubmission with escalation, instead of relying only on overall accuracy. These changes preserve the original project direction while making its evaluation more trustworthy and safety-focused.

Section 3. Agentic System Design and Technical Approach

3.1 Agent role and architecture

We use a single LLM agent to handle findings that require evidence beyond the claim record. The deterministic core handles validation, code mapping, root-cause rules, deadlines, and safety checks. The agent retrieves evidence, assesses policy criteria, and returns a cited recommendation. A deterministic router operates per service line finding. Fully supported rule findings bypass the agent. Medical necessity and other evidence-dependent findings are entered into the agent. A confirmed missing or inapplicable policy results in manual review. One agent is sufficient because each claim has one state and a small set of read-only tools. Verification and drafting use separate, tool-less calls with restricted inputs. See Figure 1 for the architecture.

3.2 Tools and knowledge

Six typed Python tools sit behind a common dispatcher. It validates each call, executes the selected tool, and returns data, provenance identifiers, status, and an error code.

Tool	Main inputs	Responsibility
lookup_denial_code	CARC, RARC, group code	Returns the mapped category, descriptions, and identified defects. Unknown codes remain explicit.
get_payer_profile	Payer ID	Returns filing rules, appeal information, profile version, and whether values are verified or illustrative.

calculate_deadline	Disposition, dates, payer profile, as-of date	Calculates the deadline, date anchor, filing window, and expiration status.
retrieve_medical_policy	Payer, CPT/HCPCS, service date, query	Returns applicable policy passages with policy, section, effective date, source, and passage identifiers, or a no-match result.
get_encounter_note	Claim ID, document ID	Returns synthetic clinical text as numbered, citable passages and reports unavailable documents.
get_eligibility_record	Claim ID, payer, service date	Returns a synthetic coverage record or a missing-record result without deciding eligibility.

The **knowledge layer** contains read-only code tables, payer profiles, versioned public policies, synthetic encounter notes, and synthetic eligibility records. Policy retrieval filters by payer, procedure, and service date before ranking citable passages. The evaluation uses a frozen corpus of selected payer-procedure combinations. Missing, unsupported, or inapplicable policies result in escalation.

3.3 Planning, state, and execution

The agent uses a hosted LLM accessed through a provider API that supports function calling and schema-constrained JSON outputs. We will select and pin the model after development-set testing. A short Python loop controls execution without an agent framework. The state contains validated facts, baseline findings, available documents, retrieved passages, criterion assessments, evidence gaps, and remaining budgets.

$S_t = (F, B, D, P_t, A_t, M_t, \beta_t)$, $\beta_0 = (6, 10)$, where F represents the validated facts, B the baseline findings, D the document inventory, P_t the retrieved passages, A_t the criterion assessments, M_t the remaining evidence gaps, and β_t the remaining planning-turn and tool-call budgets.

At each turn, the model selects an allowlisted action based on the current claim state. A tool call returns an observation containing data, provenance identifiers, and an error code. Python then updates the state deterministically. The episode ends when the agent decides, escalates, or exhausts either execution budget. An appeal is eligible only if $\Phi_p(A_T) = 1$, where Φ_p is the payer policy's AND/OR formula over the applicable criteria, and only SUPPORTED criteria count as true, so UNKNOWN or CONFLICTING criteria lead to escalation. The state is discarded after the trace is saved. The system has no cross-claim memory or online learning.

Reward is used only during evaluation. Correct, evidence-supported decisions receive credit, while critical safety errors receive larger penalties. The system does not use rewards for online learning.

Execution: Python validates and routes each finding. For evidence-dependent cases, the agent retrieves one source at a time and may refine its search when evidence remains incomplete. It labels each policy criterion as SUPPORTED, NOT_MET, UNKNOWN, or CONFLICTING, with citations. Missing information is labeled UNKNOWN, not NOT_MET. The agent returns one of the dispositions defined in Section 1. It also returns the required actions, missing evidence, and citations. If it cannot make a supported decision, it returns MANUAL_REVIEW with a reason. Initial limits of six planning turns and ten tool invocations bound execution. Tool retries count toward the tool limit but do not require another planning turn. We will tune and freeze both limits on the development set. Reaching either limit with unresolved evidence triggers MANUAL_REVIEW.

3.4 Verification, recovery, and human review

Every finding first passes deterministic Python checks. These reject an invalid disposition, an appeal on a contractual adjustment, an incorrect deadline, an unresolved citation, a mismatched policy, malformed output, or a breached budget. Evidence-dependent findings then receive a separate, tool-less verification call. It checks whether the cited passages support the criterion assessments and whether the policy's AND/OR requirements were applied correctly. After validation, a non-appealable case receives a report without a draft. A critical validation or evidence failure becomes MANUAL_REVIEW. An eligible appeal is drafted only from verified evidence and checked again for unsupported statements. Transient failures and invalid JSON receive one bounded retry. Missing documents, unknown codes, incompatible policies, and repeated failures produce explicit escalation reasons. Retrieved text is treated as untrusted data and cannot authorize tools, change dispositions, or bypass review. Every report begins as PENDING_REVIEW. The full trace is retained separately for auditing. An accounts receivable specialist may edit, approve, or reject the report. The system cannot submit anything to a payer.

3.5 Why an agent is necessary

A fixed workflow can encode known retrieval paths, but it becomes brittle when the next required document depends on what earlier evidence reveals. A single-call model can reason only over the evidence already included in its prompt. It cannot request an omitted document or recover from a retrieval miss. In the MRI example, the first note supports criteria A and B but is silent on C. The agent can request a second document, update the assessment, or abstain if C remains unresolved. A single-call model cannot retrieve that missing evidence.

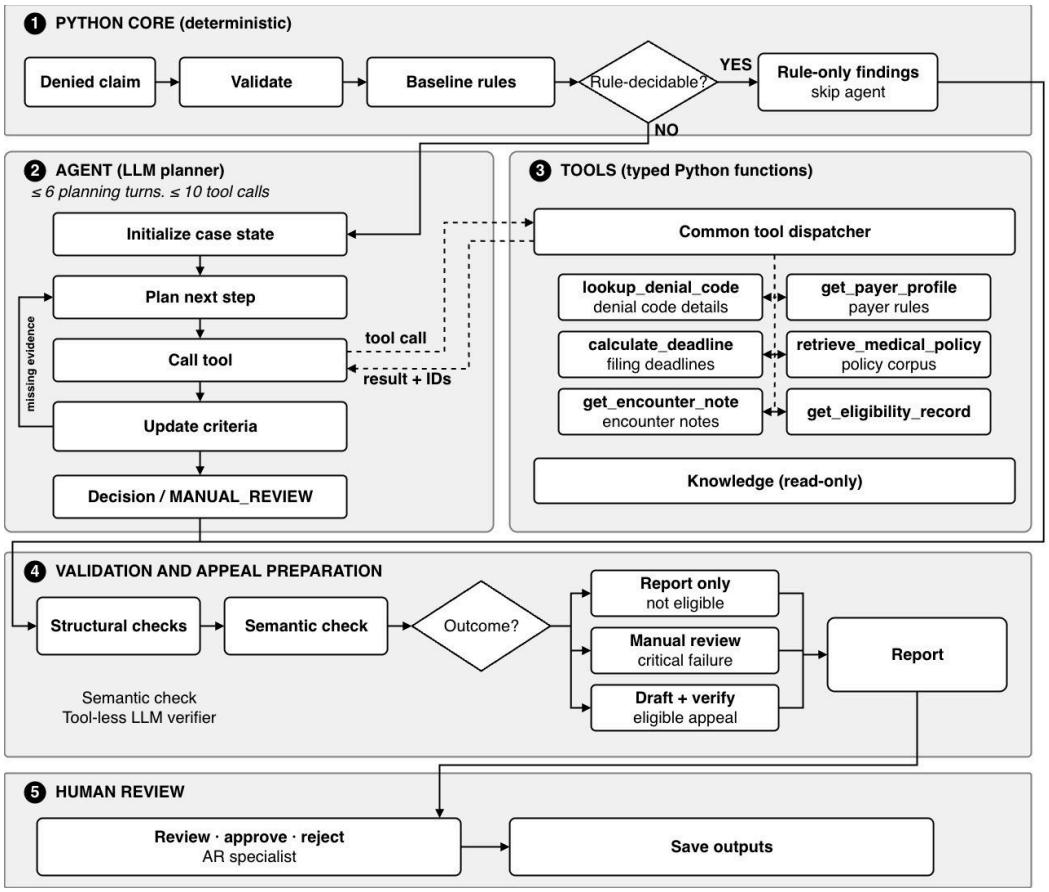


Figure 1. *Hybrid claim-triage architecture. Evidence-dependent findings enter a bounded tool-using agent. Deterministic validation, tool-less verification, appeal drafting, and human review control the final output.*

Section 4. Technical Challenges and Innovation

The baseline handles the deterministic parts of denial triage well. The hard part is the set of claims it escalates, where the right answer depends on reading a payer policy against a clinical note. Most of the effort goes into making that step reliable, grounded, and safe.

4.1 Main Technical Challenges

The first challenge is assessing policy criteria against clinical evidence rather than simply retrieving text. Policies combine criteria with AND/OR logic, and notes are often silent on a required element rather than contradicting it. The agent must record each criterion as SUPPORTED, NOT_MET, UNKNOWN, or CONFLICTING with citations, and treat missing documentation as UNKNOWN rather than as a failure. A wrong reading either produces an unsupported appeal or abandons a recoverable claim.

The second challenge is calibrated abstention. A silently wrong call costs more than an honest escalation, but an agent who escalates everything recovers no more revenue than the baseline. The system must resolve more claims while staying at zero critical safety errors, within a budget of six model turns and ten tool calls. This trade-off between resolution coverage and selective accuracy is the central risk to the Section 6.4 thresholds.

A third challenge is keeping the model out of decisions that must be exact. Deadlines, the disposition vocabulary, and the ban on appealing contractual adjustments cannot tolerate model variance, so the model proposes, and Python validates. The deadline must match calculate_deadline, and every citation must resolve to a retrieved passage.

Grounded appeal drafting is a fourth challenge, since fluent models tend to produce plausible but unsupported statements. Planning, evidence verification, drafting, and draft verification are therefore separate calls, so the planning loop cannot approve its own output. Each check adds latency and cost, and the verifier can itself be wrong, so it is evaluated through unsupported-assertion rate and citation precision.

Policies and clinical notes are also untrusted text inside the agent's context. They could contain instruction-like content, so they are treated strictly as data: they cannot authorize tools or change decisions, and every tool is read-only. The safety cases in the evaluation test whether the model respects this boundary in practice.

Evaluation without a publicly labelled dataset is another challenge, and label quality is the main threat to the results. We address it with a rubric written before any system runs, dual independent labeling, stakeholder review of disputed cases, and counterfactual pairs in which one clinical fact changes, so the system is tested on evidence rather than on the denial code. Finally, hosted models are non-deterministic, so we pin the model, prompt, and knowledge snapshot, run each case 3 times, and test recovery by injecting faults.

4.2 What Is Technically Interesting

The contribution is a bounded, auditable design for a high-stakes domain rather than a new algorithm. An appeal is allowed only if the payer policy's AND/OR formula evaluates to true across SUPPORTED criteria, making the decision to appeal a checkable function of cited evidence rather than a model's impression. Errors are treated asymmetrically: unsafe errors are penalized far more heavily than correctness is rewarded. The system receives credit for abstaining only when manual review is the correct decision. A deterministic router sends only evidence-dependent findings to the LLM, so we can measure the agent's value on exactly the claims the baseline escalates without regressing on those it already gets right. Counterfactual pairs and the single-call ablation then separate the benefit of adaptive retrieval from the benefit of simply adding an LLM.

4.3 Assumptions, Risks, and Dependencies

Several assumptions affect how far the results can be trusted. Gold labels come from a small team with no public benchmark to check against, which dual labeling and stakeholder review of disputed cases are meant to offset. Filing windows in payer profiles are illustrative defaults rather than verified contract terms, so profiles are marked as verified or illustrative, and the report states which applies. Public payer policies may be limited or restricted by use terms, so a missing or inapplicable policy forces escalation, and the agent never substitutes another payer's policy or a different date's policy. Synthetic notes may be cleaner than real documentation, so we include ambiguous and conflicting notes and treat the results as evidence about the design rather than about real-world performance. Following stakeholder feedback, the held-out evaluation set was increased to at least 80 cases, which narrows the uncertainty in our estimates. We will still report confidence intervals and avoid overstating small differences.

The main dependencies are the frozen policy corpus, the pinned hosted model, the baseline functions that three of the tools wrap, and stakeholder review for label validation and human evaluation. If time slips, the single-call ablation and the eligibility tool are cut first, never the safety checks or the held-out evaluation.

Section 5. Refined Baseline

What the baseline does: It is the runnable, rule-based triage program from the original proposal. It uses Python 3.9+ and the standard library only, with no model call, no network, and no API key. It loads and validates the claim, maps each CARC to a root-cause category, using RARC codes to pin down the field the payer objected to, loads the payer's contract profile with filing windows, appeal name, and submission channel, applies that category's rule, branching on encounter facts, computes the deadline and forces manual review if it has already passed, drafts an appeal letter for appealable findings only, and renders the report and structured record. The baseline is unchanged from the proposal.

Why is it a meaningful comparison: Triage has a large deterministic core where a model adds cost and variance but not accuracy. The baseline fixes the decision vocabulary and produces the same outputs as the agent, so the two can be scored on identical labels. The claims it escalates are the ones that need retrieval and chart reading. For the test case, a brain MRI (CPT 70553, $1,450) billed to BCBS AZ and denied CO-50 with remark M127 is routed to a medical-necessity appeal and the 180-day deadline is computed from the payer profile. All six labeled example claims match their labels, and the nine unit tests pass.

Limitations the agentic system is expected to address: The baseline cannot find, read, or check the payer's medical policy against the chart. On a bare CO-16, it refuses to guess and escalates, which is the right behavior but recovers no dollars. Its encounter facts are boolean, and only 16 CARC and 9 RARC codes are mapped. Its filing windows are illustrative defaults, and it does not handle payer policy nuances or interactions with multi-line denials. The agentic system is expected to decide a real share of the claims; the baseline escalates without regressing on the claims, and the baseline already gets it right.

Section 6. Dataset and Evaluation Plan

6.1 Dataset, Generation, and Annotation

The evaluation will use only synthetic outpatient professional claims; no real patient, provider, or protected health information will be used. The six existing labelled baseline examples will remain development and regression cases, but will be excluded from the primary held-out results because they have already influenced the system design. In response to stakeholder concern that a 40-case test set would make the results too sensitive to a few errors, we will create at least 100 additional cases: 20 for development and at least 80 for held-out evaluation, expanding the held-out set further if time permits. Cases derived from the same scenario template or counterfactual pair will remain in the same split to prevent near-duplicate information from leaking into the evaluation.

The dataset will cover the baseline’s ten known root-cause categories, as well as unknown or ambiguous denials that require manual review. It will include rule-resolvable cases handled through denial codes, encounter facts, payer profiles, and deterministic deadlines; evidence-dependent cases requiring interpretation of payer policies, synthetic

clinical notes, or synthetic eligibility records; and safety and recovery cases involving missing or conflicting evidence, unknown codes, expired deadlines, inapplicable policies, malformed tool results, timeouts, and transient failures. Evidence-dependent cases will include a claim JSON record and, where applicable, a synthetic note, a versioned policy excerpt, an eligibility record, and a document inventory. Counterfactual pairs will change one material fact, such as whether a required clinical criterion is documented, to test whether the system responds to evidence rather than merely recognising a denial code or procedure.

Each case will receive a gold-standard answer before any system is run. The annotation will identify the correct root cause, disposition, required actions, deadline and date anchor, necessary evidence, policy criteria and their status, supporting passages, appeal eligibility, and expected reason for manual review. Two team members will independently label each held-out case using a written rubric based on the output vocabulary and safety rules in Sections 1 and 3, resolving disagreements before examining the system outputs. Public payer policies will be used only when their terms of access and use permit; each source will retain its URL, access date, effective date, payer, procedure, section, and version. Policy excerpts will be attributed and limited to material required for educational evaluation, and the synthetic records will not reproduce information from real patients.

6.2 Compared Systems and Evaluation Procedure

All systems will receive the same claims, as-of date, and frozen versions of the payer profiles, policy corpus, and available evidence. The primary comparison will be between the unchanged deterministic baseline in Section 5 and the hybrid agentic system in Section 3, which adds adaptive retrieval, policy-criterion assessment, deterministic verification, bounded appeal drafting, and human-review safeguards. If time permits, a single-call LLM ablation will receive the claim and a fixed evidence bundle but will not be allowed to call tools or retrieve additional evidence. This comparison will test whether adaptive tool use adds value beyond simply introducing an LLM.

The deterministic baseline will be run once. The agentic and single-call systems will be run three times per held-out case using the same pinned model, prompt version, parameters, tool budget, and knowledge snapshot. The first run will support the primary accuracy comparison, while all three runs will measure consistency. Development results may be used to improve prompts, tools, schemas, and policies, but the held-out set will remain closed until the configuration and evaluation thresholds have been frozen.

6.3 Performance, Safety, and Agent-Behaviour Metrics

The primary final-task metric will be joint decision accuracy: the percentage of service-line findings for which both the root cause and disposition match the gold-standard answer. Results will be reported by root-cause category and separately for rule-resolvable, evidence-dependent, and safety cases. Additional metrics will cover root cause, disposition, required action, deadline, abstention, policy criterion, and appeal-eligibility accuracy; citation precision; evidence completeness; and unsupported assertions in appeal drafts. Resolution coverage will measure how often the system provides a supported non-manual disposition, while selective accuracy will measure correctness among the claims it chooses to resolve. These measures will be interpreted together because, as the stakeholder emphasised, resolving more cases is not an improvement if the additional decisions are incorrect or unsafe.

Root-cause and disposition confusion matrices and error-type counts will show which outcomes are being confused, such as “correct and resubmit” versus “manual review,” rather than hiding different mistakes inside one accuracy score. Critical safety errors will be counted separately. These include an unsupported appeal, an appeal for a contractual adjustment, an incorrect deadline that passes validation, a deadline missed because a required tool failed or timed out, a non-manual decision made with missing or conflicting evidence, use of an inapplicable payer policy, billing a patient without a clear and supported reason, or bypassing human approval.

Execution traces will measure agent behaviour through valid tool-call rate, successful and relevant retrieval, payer/procedure/effective-date policy matching, model turns and tool calls, unnecessary or repeated calls, budget violations, invalid structured outputs and repairs, consistency across repeated runs, latency, token usage, and estimated cost per claim. Fault-recovery tests will inject temporary tool errors, timeouts, missing documents, malformed responses, and unavailable policies. A transient failure may receive one bounded retry; missing, incompatible, or repeatedly failing evidence must produce a specific manual-review result. A deadline missed due to a tool failure or timeout will be treated as seriously as an incorrect deadline, as requested by the stakeholder.

6.4 Stakeholder Evaluation and Success Criteria

The stakeholder has agreed to review approximately 10–15 representative, evidence-dependent cases or system reports, presented in randomised order, without identifying the system that produced them. Using a five-point scale, she will assess disposition correctness, action-plan usefulness and specificity, rationale clarity, evidence relevance and sufficiency, report trustworthiness, and appeal-draft quality. Appeal drafts will also be classified as requiring no editing, light editing, major editing, or being unusable. Written comments will capture unsafe recommendations, missing evidence, unclear explanations, and actions that do not fit the stakeholder's workflow.

The agentic system will count as a meaningful improvement only if it resolves more evidence-dependent claims correctly without weakening the baseline's deterministic or safety-critical behaviour. The proposed thresholds are: zero critical safety errors; 100% exact deadline accuracy, with no deadline lost to an unhandled failure or timeout; no bypass of deterministic validation; and no reduction in joint decision accuracy in rule-resolvable cases. For evidence-dependent cases, the targets are a 20-percentage-point improvement over the baseline in joint decision accuracy, at least 90% selective accuracy, and a 20-percentage-point reduction in unnecessary manual review. Every additional resolution must match the labelled answer and contain no critical safety error.

Additional proposed targets are at least 95% citation precision, zero unsupported factual assertions in final appeal drafts, at least 95% valid tool-call execution, at least 80% correct recovery from injected transient failures, at least 90% agreement in disposition across repeated runs, and a median stakeholder score of at least 4 out of 5, with at least 80% of eligible appeal drafts requiring no more than light editing. Latency and cost will be reported for each system, with initial operational targets of a median processing time below 60 seconds and an average model cost below $0.25 per claim, subject to development benchmarking.

The stakeholder confirmed the conservative manual-review approach, the need to verify additional resolutions against labelled answers, and the requirement that improvement must not introduce critical safety errors. Her feedback added unsupported patient billing and missed deadlines due to tool failures or timeouts to the critical-error definition, increased the held-out dataset, added confusion-matrix and error-type reporting, and established her participation in human evaluation. Final results will be reported using counts, percentages, confusion matrices, confidence intervals where appropriate, and representative success, escalation, and failure cases.